

METADV: A Unified and Interactive Adversarial Testing Platform for Autonomous Driving

Aishan Liu
Beihang University, Beijing, China
liuaishan@buaa.edu.cn

Jiakai Wang
Zhongguancun Laboratory, Beijing, China
wangjk@zgclab.edu.cn

Tianyuan Zhang
Beihang University, Beijing, China
zhangtianyuan@buaa.edu.cn

Hainan Li
Institute of Dataspace, Hefei, China
hainan@buaa.edu.cn

Jiangfan Liu
Beihang University, Beijing, China
liujiangfan@buaa.edu.cn

Siyuan Liang
Nanyang Technological University, Singapore, Singapore
pandaliang521@gmail.com

Yilong Ren
Beihang University, Beijing, China
yilongren@buaa.edu.cn

Xianglong Liu†
Beihang University, Beijing, China
Zhongguancun Laboratory, Beijing, China
Institute of Dataspace, Hefei, China
xlliu@buaa.edu.cn

Dacheng Tao
Nanyang Technological University, Singapore, Singapore
dacheng.tao@ntu.edu.sg

Abstract

Evaluating and ensuring the adversarial robustness of autonomous driving (AD) systems is a critical and unresolved challenge. This paper introduces METADV, a novel adversarial testing platform that enables realistic, dynamic, and interactive evaluation by tightly integrating virtual simulation with physical vehicle feedback. At its core, METADV establishes a hybrid virtual-physical sandbox, within which we design a three-layer closed-loop testing environment with dynamic adversarial test evolution. This architecture facilitates end-to-end adversarial evaluation, ranging from high-level unified adversarial generation, through mid-level simulation-based interaction, to low-level execution on physical vehicles. Additionally, METADV supports a broad spectrum of AD tasks, algorithmic paradigms (e.g., modular deep learning pipelines, end-to-end learning, vision-language models). It supports flexible 3D vehicle modeling and seamless transitions between simulated and physical environments, with built-in compatibility for commercial platforms such as Apollo and Tesla. A key feature of METADV is its human-in-the-loop capability: besides flexible environmental configuration for more customized evaluation, it enables real-time capture of physiological signals and behavioral feedback from drivers, offering new insights into human-machine trust under adversarial conditions. We believe METADV can offer a scalable and unified framework for adversarial assessment, paving the way for safer AD. Our demo can be found at <https://sites.google.com/view/metadv-demo-video>.

†The Corresponding author.

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1 Introduction

Autonomous driving has made significant progress in recent years [2, 6, 13, 17, 18, 20, 38, 39], yet remain highly susceptible to *adversarial attacks* [10, 31], ranging from digital perturbations in sensor input [4, 11] to physical-world camouflage [7, 22, 36], which can lead to critical failures and safety hazards. As these systems move closer to widespread deployment, evaluating and ensuring their adversarial robustness has become an urgent and unresolved challenge. However, current adversarial testing approaches are hampered by the lack of a unified, interactive, and dynamic evaluation platform. Existing studies [19, 37] often focus narrowly on isolated and static environments, limiting their capacity to uncover the full spectrum of weaknesses that may emerge in real-world conditions.

To address this, we present METADV, a novel, unified, and interactive adversarial testing platform for AD (shown in Fig. 1). Central to METADV is a hybrid virtual-physical sandbox built on CARLA, which enables real-time, closed-loop adversarial evaluation from virtual simulation to physical deployment. METADV features a three-layer hierarchical architecture, unifying the high-level adversarial attack generation and connecting mid-level simulation operations to low-level physical execution. Here, we also propose dynamic adversarial testing evolution that can generate targeted adversarial test cases to better reveal threats covering the full perception, prediction, and planning pipeline. METADV is further distinguished by its integration of a rich AD asset repository, encompassing a broad spectrum of tasks, algorithmic paradigms, and 3D vehicle models. It also supports seamless transitions from simulation to deployment,

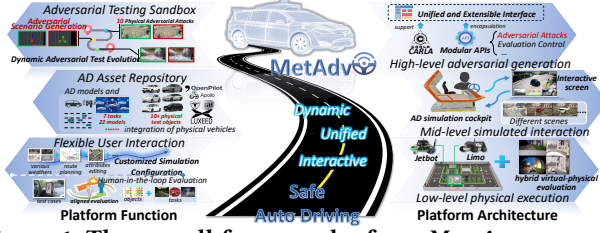


Figure 1: The overall framework of our METAdv, a novel virtual-physical hybrid adversarial testing platform for AD.

with plug-and-play compatibility for commercial platforms such as Apollo [8, 9] and Tesla [32]. Besides, METAdv offers extensive user interactivity, allowing real-time editing of environments, adversarial artifact insertion; crucially, it supports human-in-the-loop evaluation: by capturing real-time physiological signals and behavioral feedback (e.g., electroencephalography), it enables the study of human-machine trust under adversarial conditions. METAdv supports software-in-the-loop (SIL), hardware-in-the-loop (HIL), and vehicle-in-the-loop (VIL) testing, which serve as a diagnostic tool to expose hidden vulnerabilities for more trustworthy AD vehicles.

2 METAdv System

2.1 Adversarial Testing Sandbox

We first propose a unified closed-loop evaluation framework that enables the real-time adversarial testing and bridges the gap between simulation and reality; we then propose the dynamic adversarial testing evolution that generate adversarial cases targeted to AD.

Unified Closed-loop Evaluation. The overall framework of our system consists of three hierarchical layers: ① *High-level Unified API Layer*. Built upon CARLA 0.9.15 [5], this layer provides a unified and extensible interface for adversarial evaluation. We encapsulate core functionalities (e.g., adversarial attacks, evaluation control) into modular APIs to enable standardized and reusable testing workflows. ② *Mid-level Simulation Operation Layer*. This layer manages interaction with the simulator, including scenario deployment, vehicle control, and sensor emulation. It supports precise control over environmental dynamics and agent behaviors. ③ *Low-level Physical Execution Layer*. This layer enables the connection of simulation control commands to real-world physical vehicles via hardware integration in a faithfully reproduced setting. This framework enables closed-loop evaluation in the interactive environment, supporting SIL, HIL, and VIL level testing.

Dynamic Adversarial Test Evolution. Based on the evaluation environment, we then propose to generate adversarial test cases targeted at the test-taking AD. In particular, METAdv can generate dynamic *adversarial scenario* to the AD to impact their decision: given a initial benign scene, we use a large language model (LLM) to infer an adversarial agent whose behavior poses a threat to the ego AD; subsequently, we perform scenario evolution by introducing background vehicles with collaborative risky trajectories to increase the likelihood of collision. Based on the adversarial scenario, we can also perform targeted *digital* and *physical adversarial attacks* to the ego AD. Our system features 23 prominent digital adversarial attacks (e.g., FGSM [10], PGD [25]) to rigorously evaluate the robustness of perception modules in ADs by adding imperceptible perturbations to raw sensor data. Besides, METAdv has implemented 10 physical adversarial attacks, e.g., DAS [36]

and FCA [35], that applying adversarial camouflage onto physical objects, using a novel dual-renderer fusion strategy (a differentiable renderer to simulate the adversarial textures and a native CARLA renderer to preserving the unperturbed background).

2.2 AD Asset Repository

AD Models and Tasks. METAdv supports a diverse range of AD algorithms covering perception, prediction, and planning stages, and includes 7 classical AD tasks (e.g., obstacle recognition, scene understanding). Specifically, METAdv consists of 10 deep learning based perception models (e.g., YOLOs [28], SMOKE [23]), 5 reinforcement learning based decision AD models (e.g., DDPG [21], PPO [29]). Additionally, we incorporate unified frameworks that cover the full pipeline, including 4 end-to-end models (e.g., UniAD [12], VAD [17]) and 3 LLM-based AD models (e.g., LMDrive [30], Dolphins [24]). We also introduce 6 high-quality 3D vehicle models represented in .obj files, such as Tesla Model 3.

Integration of Physical Vehicles. Besides the SIL test, we enable our system with HIL and VIL tests and endow METAdv with a progressive testing ability for the integration of physical hardware and vehicles to achieve the practical AD adversarial testing. Benefited from the integrated progressive verification system, METAdv has provided the adversarial testing ability on 2 AD platforms (Apollo [8, 9], OpenPilot [1]), 2 laboratory vehicles (LIMO [27], Jetbot [26]), and 5 commercial vehicles (Tesla Model 3/Y [33, 34], XIAOMI SU7 [3], XPENG G6/P7+ [15, 16], and LUXEED [14]).

2.3 Flexible User Interaction

Customized Simulation Configuration. METAdv offers a highly customizable environment for evaluating ADs via a set of unified interfaces. Environmental settings such as cloud density, solar position, and fog concentration can be precisely adjusted to emulate various weathers; route planning is supported through the specification of start and end coordinates with a diverse set of map layouts; vehicle attributes, including type and texture, are also configurable to simulate variability in visual appearance; attack methods, AD models, and tasks can be easily configured. Our system also supports the easy extension of new attacks, algorithms, and maps to better simulate the real-world AD conditions.

Human-in-the-loop Evaluation. METAdv also involves human-in-the-loop evaluation and considers the more practical scenario where humans and machines work collaboratively in the real world. Specifically, METAdv collects real-time human feedback (i.e., electroencephalography, eye movement) and enables drivers participating in the whole driving process for a trustworthy AD evaluation at both HIL and VIL. Specifically, at HIL level, METAdv is been deployed on an AD simulation cockpit which allows the users to participate in the test flow and interrupt the operation of the tested AD system, thus providing a human-machine aligned perspective study; at VIL level, METAdv offers a real-vehicle monitoring environment to provide a more humane decision-making evaluation dimension. Every takeover by human drivers could be recorded in this environment, allowing users to intuitively judge the adversarial robustness of an AD system by the comparative analysis of takeover frequencies. This design has great potential to be explored, especially for AD testing of human-machine mutual trust.

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